INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, NAGPUR

Department of Computer Science and Engineering

Dr. Jagdish Chakole

Computational Finance

Reinforcement Learning for Stock Market Trading Strategy using Q-Learning

GROUP 8:
BT21CSE122 Vikash Rajput
BT21CSE123 Harsh Lohiya
BT21CSE141 Uday Bhati
BT21CSE142 Vyankatesh Deshpande
BT21CSE148 Prashant Kumar

Table Of Contents

1.

1. Objective

Develop a trading strategy using a Reinforcement Learning

Train an RL agent to learn an optimal trading policy

Use historical price data and technical indicators as inputs

Choose a trading style: intraday, swing, or long-term investing

2. Data Collection & Preprocessing

Market Selection & Data Configuration

- Data Source: Yahoo Finance API (yfinance)
- Sample Ticker: AAPL (Apple Inc.)
- Timeframe Options: Daily
- Custom Date Range: Configurable using start_date and end_date parameters

Data Acquisition Method

- Source: Yahoo Finance API via yfinance library
- Implementation: Handled in load_data function.

Technical Indicators Implemented

- Trend Indicators:
  - SMA (20 & 50)

- Momentum Indicators:
  - RSI

    Overbought/Oversold Signals

- Data Normalization
  - Purpose: Scales numeric features between 0 and 1 for model efficiency

- Data Splitting
  - Split Type: Temporal (time-based) split
  - Default Ratio: 80% training, 20% testing
  - Implementation: Handled in split_data function.

3. Reinforcement Learning Components

Reinforcement Learning Components – State Space

- Technical Indicators:
  - Trend: SMA (20, 50)
  - Momentum: RSI
  - Close Price
- Account Info:
  - Current balance
  - Shares held
  - Current stock price

Reinforcement Learning Components – Action Space

- Action Space Type: Discrete
- Available Actions:
  - Buy – Purchase stock
  - Sell – Sell held stock
  - Hold – Take no action

Reinforcement Learning Components – Reward Function

- Profit-Based Reward: Based on % change in portfolio value
- Indicator Alignment: Bonus for trades matching technical signals
- Drawdown Management: Penalty if portfolio drops >5%
- Inactivity Penalty: Discourages holding without action

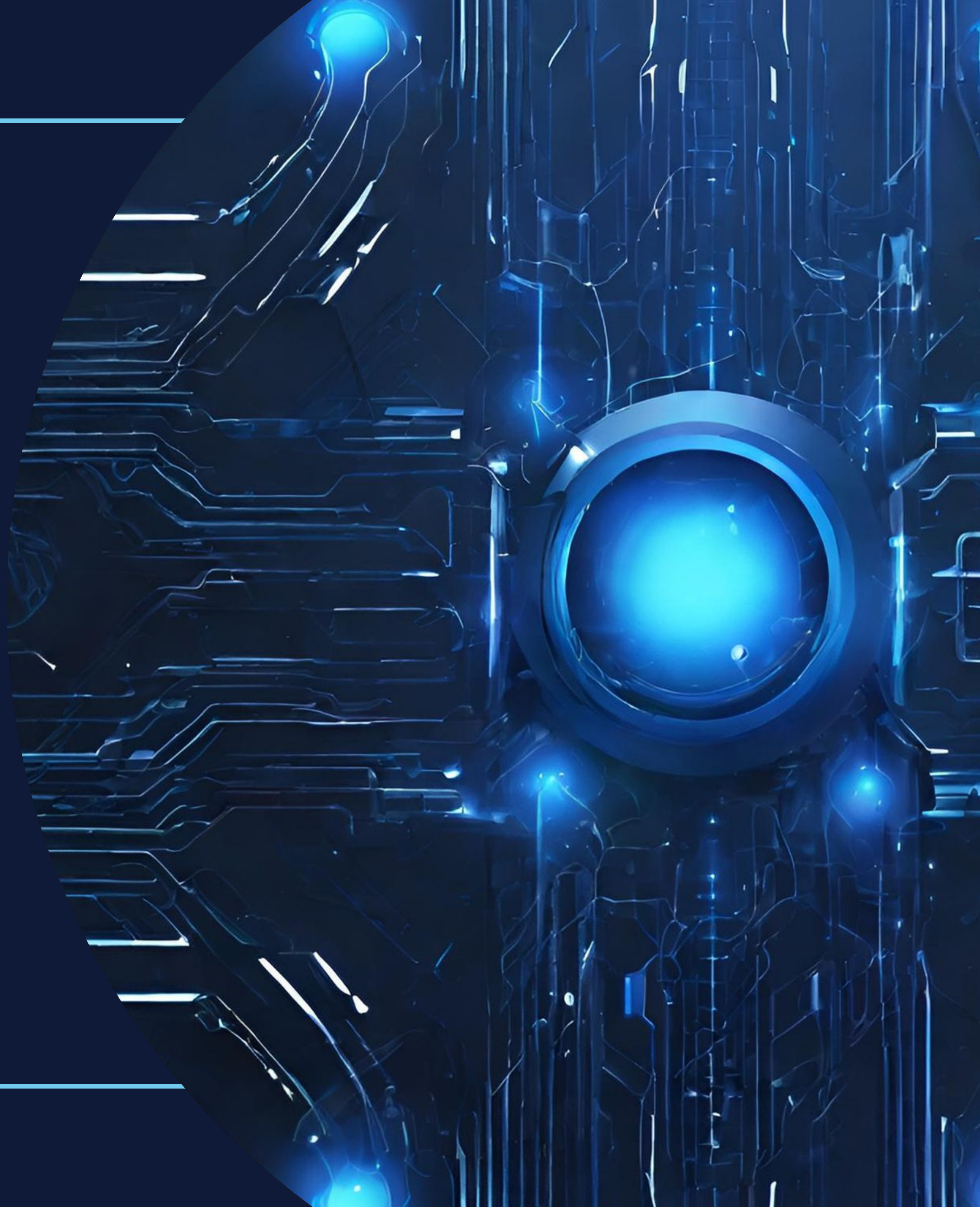
4. Reinforcement Learning Algorithm – Q-Learning

- Approach: Tabular Q-Learning
- Key Components:
  - Q-table to store state-action values
  - Epsilon-greedy policy for exploration vs. exploitation
  - Learning rate (α), discount factor (γ), and exploration rate (ε)
  - Q-value update based on Bellman Equation

5

Implementation

Implementation of RL model
with q_learning agent



- Q-Table Definition
  - Data Structure: Nested defaultdict
  - State-Action Mapping: Discretized states mapped to action values
  - Default Value: 0 for new state-action pairs
  - Handling Continuity: Continuous states discretized for Q-table use

- Bellman Equation Implementation
  - Q-Learning Update Rule:
  - Follows the standard Bellman equation:
  - $Q(s,a)=Q(s,a)+\alpha[r+\gamma a' \max Q(s',a')-Q(s,a)]$
  - Application: Used in q_learning_a to update Q-values based on observed rewards and future values

- State Space Handling
  - Discretization: Continuous state space is discretized for Q-table mapping
  - State Representation:
    - Technical indicators, account info, and stock price discretized into discrete bins
    - Each discretized state is used as a key in the Q-table

- Action Selection (Epsilon-Greedy)
  - Exploration vs. Exploitation: Uses an epsilon-greedy policy
  - Exploration Rate Decay: Gradual reduction in exploration (ε) over time
  - High ε: Encourages exploration in early stages
  - Low ε: Prioritizes exploitation as the agent learns

- Action Choice:
  - Explore: Random action with probability ε
  - Exploit: Best-known action with probability $1 - \varepsilon$

6. Training and Evaluation

Train and Evaluate the Model – Training Process

- **Data Split:** Dataset divided into training and testing sets

- **Training Approach:**
  - Agent learns by maximizing rewards across multiple episodes
  - Exploration rate (ε) gradually decays to shift focus from exploration to exploitation

- **Goal:** Optimize the agent's policy through repeated interactions with the environment

Evaluation Metrics

- **Metrics Calculated:**
  - **Total Profit:** Overall portfolio gain or loss
  - **Sharpe Ratio:** Risk-adjusted return measurement
  - **Max Drawdown:** Largest drop in portfolio value
  - **Win Rate:** Percentage of profitable trades
  - **Cumulative Return:** Total return over time

Baseline Comparison

- **Comparison with:**
  - **Moving Average Strategy:** Trades based on predefined moving average crossovers

Visualization of Results

- **Training Progress:**
  - Graphs showing rewards, portfolio values
- **Equity Curves:**
  - Comparison of RL strategy, Moving Average Crossover, and Buy & Hold strategies

All metrics and results are saved further analysis.

7. Visualizations and Report

Visualizations

- **Equity Curve:**
  - **RL Strategy:** Solid blue line, final return: **7.18%**
  - **MA Crossover:** Orange dashed line, return: **2.08%**

- **Trading Signals on Price Chart:**
  - AAPL price with SMA 20 & SMA 50 (green & orange dashed lines)
  - **Buy signals:** Green triangles | **Sell signals:** Red triangles
  - Highlights entry/exit points, concentrated early in test period

- **Q-Learning Progress:**
  - **Training Rewards:** Upward trend over episodes
  - **Portfolio Value:** Grew from $10,000 with volatility
  - **Exploration Rate Decay:** Gradual drop from **1.0 to ~0.6** over 50 episodes

Performance Metrics Comparison

- **Total Return:**
  - RL > MA Crossover
- **Sharpe Ratio:**
  - RL outperforms MA Crossover (better risk-adjusted returns)

Report Components Overview

- **Model & Training:**
  - **Q-Learning:** Tabular approach with discretized state space
  - **State Space:** Combines price data, technical indicators, and account info
  - **Action Space:** Discrete — Buy, Sell, Hold
- **Training Process:**
  - Starts with full exploration ($\varepsilon = 1.0$), gradually shifts to exploitation
  - **Rewards:** For profitable trades & indicator-aligned actions
  - **Penalties:** For overtrading & significant drawdowns (>5%)
  - **Q-Value Update:** Bellman equation
    - **Learning rate (α):** 0.001
    - **Discount factor (γ):** 0.95

Results Discussion

- **Relative Performance:**
  - RL Strategy Return:
    ((10718 - 10000) / 10000) * 100% ≈ +7.18%

  - MA Strategy Return:
    ((10208 - 10000) / 10000) * 100% ≈ +2.08%

- **Risk Management & Trading Behavior:**
  - RL executed more trades early in the test period
  - More dynamic than MA Crossover in response to market shifts

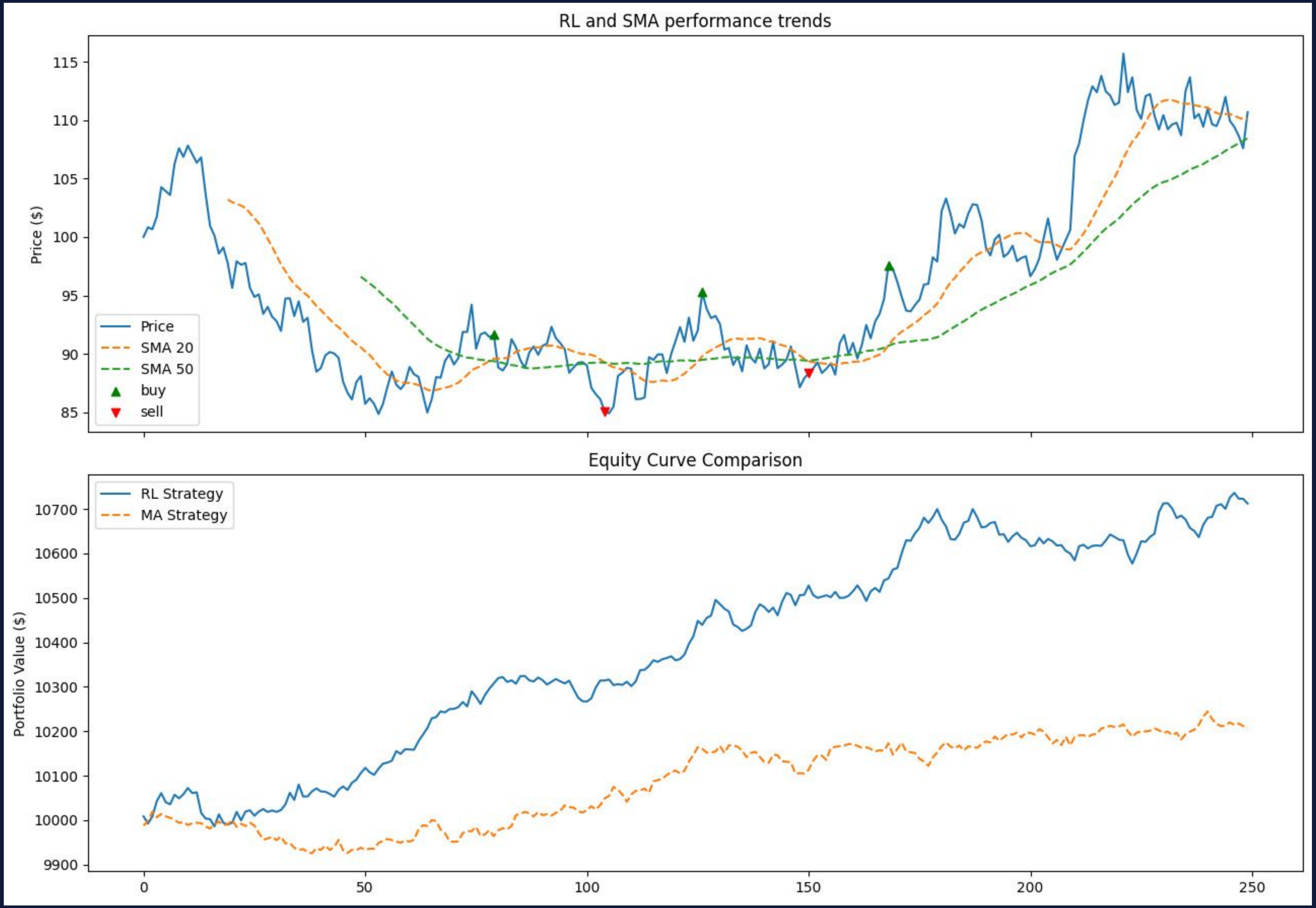
Strengths & Limitations

- **Strengths:**
  - Adaptive, market-responsive decision-making
  - Utilizes multiple indicators for robust signals
- **Limitations:**
  - Struggles in strong bear markets
  - Sensitive to hyperparameters & training duration

Comparison with Baseline Strategies

- **RL vs. MA Crossover:** 5.**1%** higher return
- **Trading Activity:** RL was more active than MA Crossover

Actions & Equity curve



Thank you!